

Use Mahout for Clustering Big Data

Various techniques are used for scaling Big Data to make it comprehensible and intelligible. Clustering is a technique to arrange groups of objects into classes of similar objects. This article outlines a technique for clustering in Mahout, which is a library of scalable machine-learning algorithms.



Mahout is an open source machine learning library from the Apache Software Foundation. It implements many data mining algorithms like *Recommender engines()*, *clustering()*, *classification()* and is scalable to very large data sets (up to terabytes and petabytes) that are in the Big Data realm.

This article explains clustering techniques in Mahout. Clustering is used to arrange a group of abstract objects into classes of similar objects. It basically works on collection of data that is unlabelled and then analysed to identify a pattern, so that the data can be grouped together. Thus clustering discovers groups of similar features together. Clustering techniques are used in many fields such as market research to discover different groups of customers, in image processing for segmentation of images and pattern recognition, and in text or document classification for information discovery. These techniques are also used in outlier detection (also called anomaly detection), which means identifying events or observations that do not conform to an expected outcome, to help in identifying fraud in online transactions, etc.

The Clustering algorithms implemented in Apache Mahout are K-Means, Fuzzy K-Means, Streaming K-Means and Spectral Clustering. Clustering a group of objects involves three things:

- An algorithm, which is the method used to group things together.
- A notion of both similarity and dissimilarity — which item belongs to an existing stack and which should start a new one.
- A stopping condition, which might be the point beyond which objects can't be stacked any more, or when the stacks are already quite dissimilar.

I will explain the clustering of news articles using K-means clustering, using Mahout on top of Hadoop. In this example, similar types of news articles will be grouped together. Hence, all news articles related to politics will be grouped together, those related to sports will be grouped together, and so on. The K-means clustering algorithm partitions 'n' observations into 'k' clusters, in which each observation belongs to the cluster with the nearest mean.

There are three steps involved in inputting data for the Mahout clustering algorithms. These are as follows.

- Identify various news articles, paste them in different text files in a single folder, and copy this folder to a Hadoop file system using the following command:

```
hadoop fs - put /home/hduser/news_article /news_article
                Source                               Destination
```

- Convert the files to sequenced files using the command *seqdirectory*, which generates an intermediate document